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Report on the Data Analysis

Research Question and Hypothesis

Research question: Can regions with relatively few accommodation companies be more specialized in this activity than regions with many such companies?

Hypothesis: Rankings based on the number of companies and on the share of an activity within the regional business structure may differ substantially. To test this hypothesis, I compared absolute counts, regional shares, and Location Quotients across Chile's 16 regions.

Data and Preparation

The original dataset contains 476,615 unique companies. NAICS classifications are available for 442,374 of them, while the address data include both regional names and codes. Regional names sometimes appear in different language variants, so I used standardized regional codes for Chile's 16 regions.

I extracted the NAICS classification and its associated score from the nested data structures, validated the regional codes, and identified companies operating in multiple regions. In the extended analysis, a company may contribute to several regions, but only once per company–region pair. After filtering for classified companies with valid regions, 386,136 unique companies and 409,828 company–region relationships remained. No duplicate pairs were found.

Accommodation was defined using the NAICS 721 prefix. This covers accommodation, not the entire tourism industry or food services (NAICS 722). The final dataset identified 22,034 unique companies classified under NAICS 721; the number of regional presences may be higher.

Method Used to Measure Specialization

Regional share = number of NAICS 721 company presences in a region / total number of classified company presences in that region.

Reference share = number of NAICS 721 company presences across all regions / total number of classified company presences across all regions. In the final analysis: approximately 5.6519%.

Location Quotient (LQ) = regional share / reference share. An LQ greater than 1 means the activity is overrepresented in that region relative to the reference sample; an LQ below 1 indicates

underrepresentation. LQ does not measure revenue, employment, productivity, or contribution to GDP.

Main Finding (Volume versus Specialization)

Indicator	Santiago (cl-rm)	Aysén (cl-ai)
NAICS 721 presences	4,171	737
Total classified presences	191,941	3,537
NAICS 721 share	2.17%	20.84%
LQ	0.38	3.69

As the table shows, Santiago has more than five times as many accommodation company presences as Aysén. However, Aysén has a much higher regional share. This contrast brings the central message into focus: absolute volume and relative specialization are not the same thing.

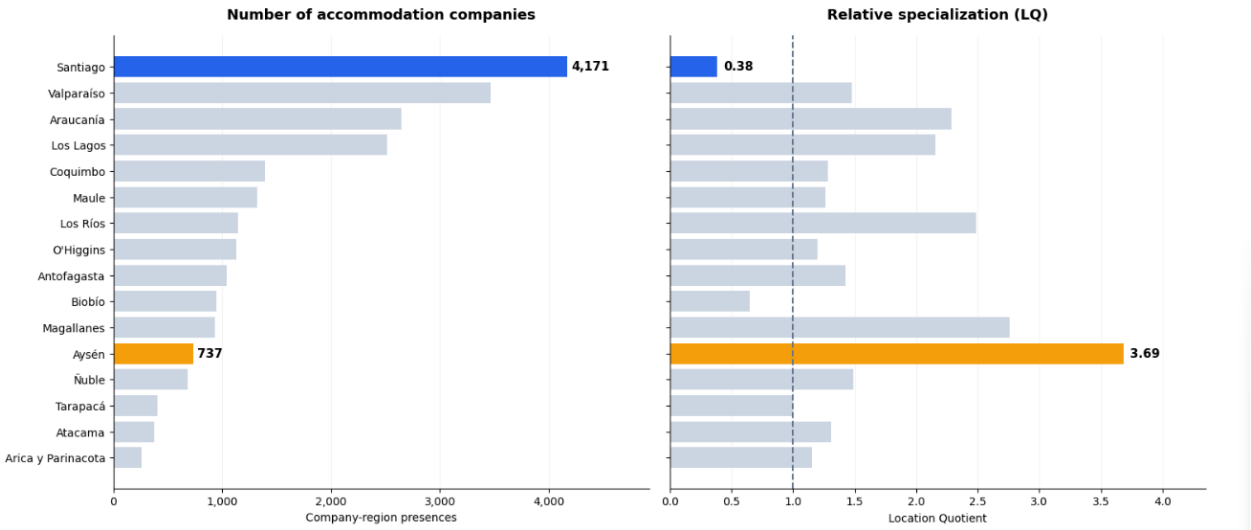


Figure 1. Comparison of accommodation company presences and LQ across the 16 regions.
Source: Author's analysis of Veridion Chile data

Robustness Testing

I recalculated the indicator under three NAICS score thresholds: no additional filter, score ≥ 0.5, and score ≥ 0.8. For each threshold, the reference share was recalculated using the corresponding sample; therefore, LQ values across scenarios should not be interpreted as changes over time.

Region	Threshold 0	Threshold 0.5	Threshold 0.8

Aysén	3.687	4.071	4.055
Magallanes	2.760	2.953	3.389
Los Ríos	2.484	2.850	2.257
Araucanía	2.285	2.688	2.709
Los Lagos	2.159	2.327	2.054

Aysén ranks first in all three scenarios. Other regions change positions and LQ values, indicating some sensitivity to the threshold even though the main finding persists. The number of unique companies falls from 386,136 to 244,061 and 155,986 as the filter becomes stricter.

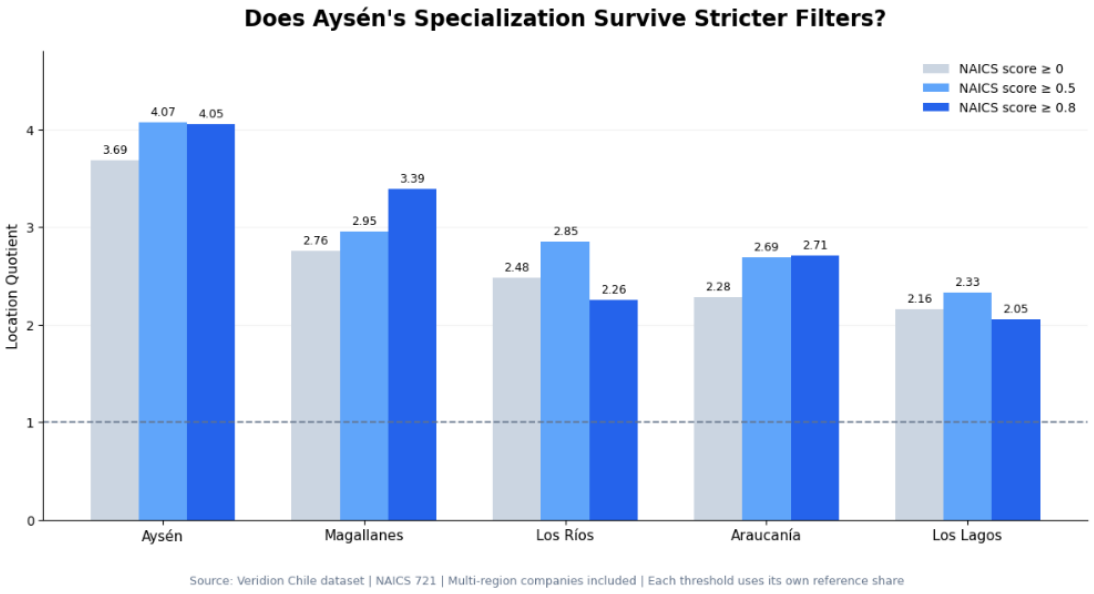


Figure 2. LQ robustness test for the five highest-ranked regions in the baseline scenario.

Quality Checks

As a final quality check, I conducted an audit confirming 409,828 company–region relationships and 386,136 unique companies; zero duplicate pairs; regional codes belonging to the 16 Chilean regions; zero mismatches between the NAICS 721 rule and the accommodation label; a negligible numerical difference when recalculating LQ (approximately 4.44×10^{-16}); and Aysén ranking first at every tested threshold. These checks confirm the internal consistency of the implementation.

Limitations and Alternative Interpretations

The Veridion data are not presented here as a complete census of companies in Chile. Coverage and classification likelihood may vary across regions and sectors. Companies with multiple locations are counted in each valid region, and a company's presence does not indicate the scale of its operations in that region.

A high LQ may reflect a different local economic profile, not necessarily high tourism activity. I did not use data on overnight stays, tourist arrivals, revenue, or employment; therefore, the explanation that “tourism drives specialization” remains a plausible hypothesis rather than a demonstrated causal relationship.

Filtering by score may change both the numerator and the denominator and may favor certain types of companies. Consequently, the robustness test supports the stability of the ranking under the selected scenarios, not the external validity of the entire dataset.

Conclusion

The analysis shows why a regional comparison based solely on company counts can be misleading. Santiago has the largest number of accommodation company presences in the analyzed dataset, but Aysén has the highest relative specialization. This finding remains stable across the tested NAICS thresholds and illustrates the value of combining absolute counts with normalized indicators and methodological checks.